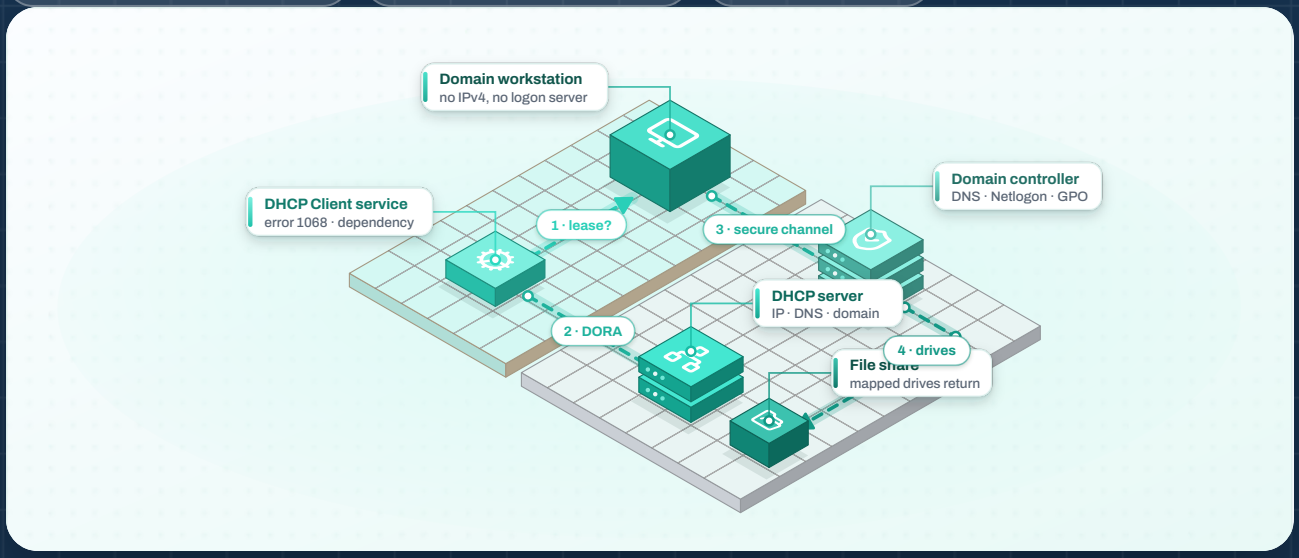
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-WIN-002

Restoring the DHCP Client Service and Domain Connectivity

Bring a workstation back from "no network, no domain" four ways: the Services and Event Viewer consoles, sc/ipconfig/nltest at the command line, a PowerShell repair template, and a controlled domain rejoin, plus Intune detection at scale.

Windows Desktop Support | Networking and Services | Active Directory



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Sanitized reference edition — client identifiers removed

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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain how DHCP leases an address (DORA) and why every later step of a domain logon depends on it.
- Read and change Windows service configuration: start type, svchost host groups and service dependencies.
- Use sc qc, sc config depend= and Get-Service -RequiredServices safely, including restoring DHCP Client's default dependencies (NSI, Tcpip, Afd).
- Diagnose and repair a broken computer secure channel without rejoining the domain where possible.
- Verify recovery with ipconfig, nltest, gpupdate/gpresult and the System event log.
- Detect dependency drift across a fleet with an Intune Remediation.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
15–30 min (service) · +15 min (rejoin)	Intermediate	Local admin; delegated domain rights to reset/rejoin	Moderate: service config and domain membership

CHAPTER 1

Concepts You Need First

A domain-joined workstation that "has no network" usually produces a cascade of symptoms: no internet, "There are currently no logon servers available", Group Policy errors, missing mapped drives, and sometimes "The trust relationship between this workstation and the primary domain failed". It is tempting to start at the end of that list. This SOP starts at the beginning, because in the reference engagement every one of those symptoms came from a single root cause: the **DHCP Client** service could not start because of a broken **service dependency**.

KEY CONCEPT · A DOMAIN LOGON IS A RELAY RACE

Think of reaching a mapped drive as a relay race with seven runners. The network card hands the baton to DHCP, DHCP to DNS, DNS to the domain controller locator, then the secure channel, Group Policy and finally the logon script that maps drives. If the second runner never starts, the five after it never receive the baton, and each of them reports its own failure. Fix the first runner that dropped the baton and the rest usually recover on their own.

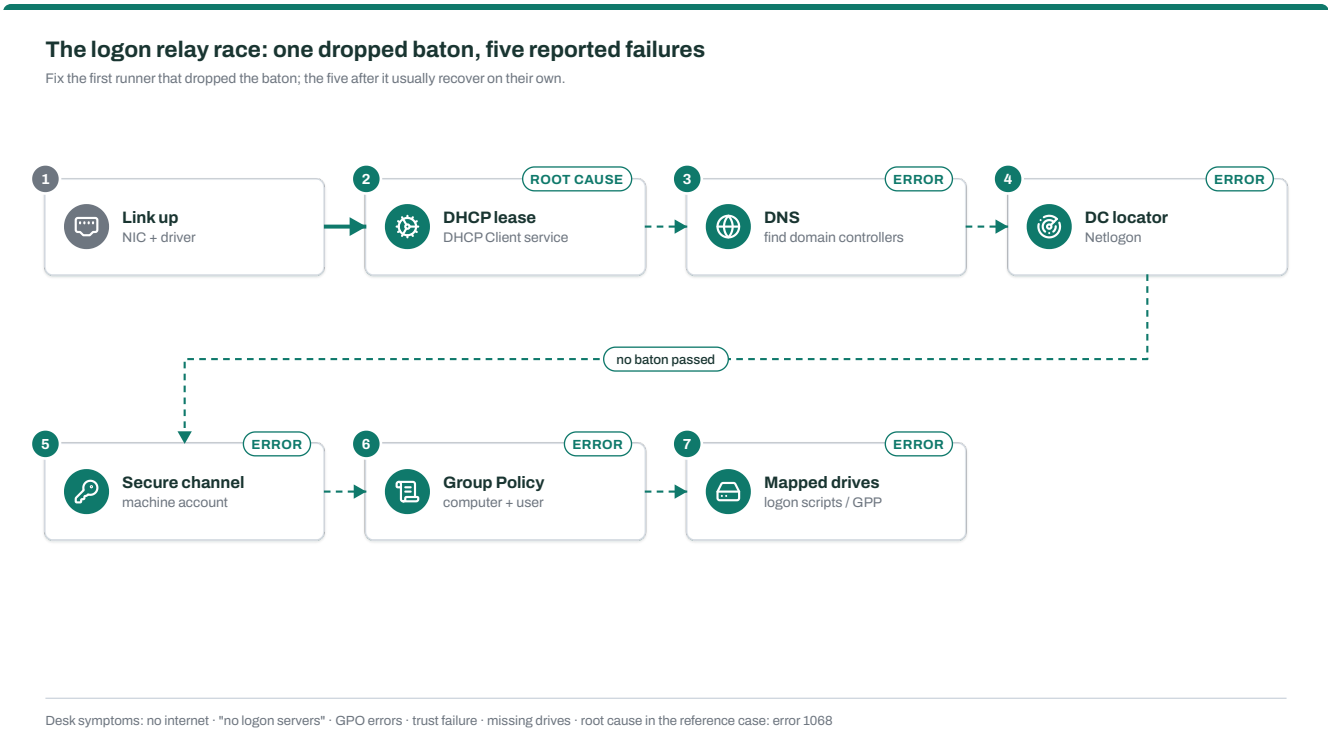


Figure 1. Start at the first link in the chain that broke; the errors further along are usually symptoms, not separate faults.

1.1 How DHCP gives a workstation its address

The **Dynamic Host Configuration Protocol** (DHCP) leases an IP configuration to a client for a limited time. The exchange has four messages, remembered as **DORA**: Discover, Offer, Request, Acknowledge. The lease carries far more than an IP address: subnet mask, default gateway, DNS servers and DNS domain name all arrive in DHCP *options*. Without DNS servers, a domain member cannot find a domain controller.

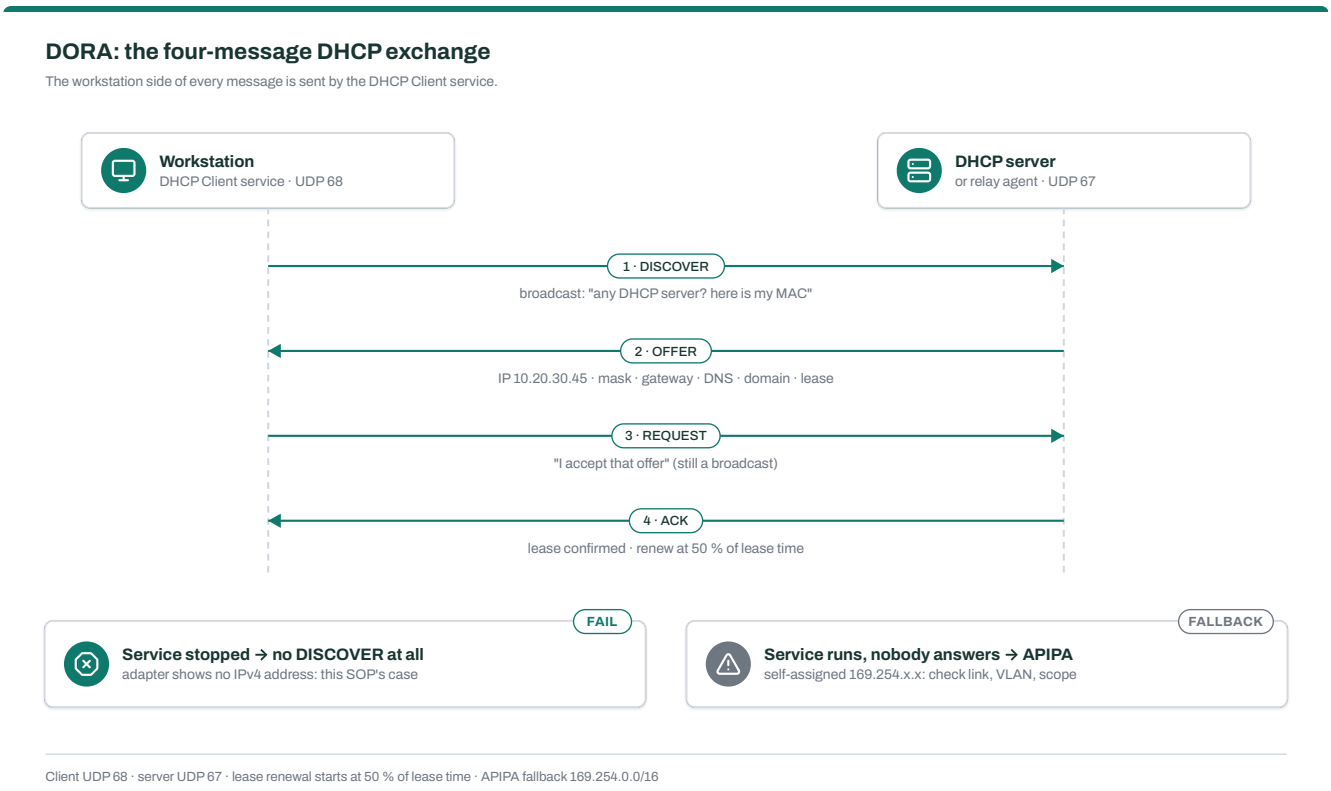


Figure 2. A workstation with no IPv4 address at all never sent DISCOVER, which points at the DHCP Client service rather than the network.

On Windows, the client side of DORA is performed by the **DHCP Client** service (service name `Dhcp`). It also renews leases, registers the computer's A and PTR records in DNS (dynamic DNS), and assigns an **APIPA** address (169.254.x.x) when no DHCP server answers. That last point is diagnostic gold:

What ipconfig shows	What it means
A normal address, e.g. 10.20.30.45	DHCP works. Look further up the chain (DNS, secure channel).
169.254.x.x (Autoconfiguration IPv4)	The service is running but no DHCP server answered: cable, VLAN, switch port, 802.1X or scope exhaustion.
No IPv4 address at all on a connected adapter	The DHCP Client service is not running. This is the case this SOP repairs.

1.2 Windows services, start types and svchost groups

A **service** is a program that the **Service Control Manager** (SCM, `services.exe`) starts and supervises outside any user session. Each service's configuration lives in the registry under `HKLM\SYSTEM\CurrentControlSet\Services\<name>`: its start type, the account it runs as, its binary path and its dependencies.

Many Windows services are not separate programs but DLLs hosted inside a shared **service host** process, `svchost.exe`. The `-k` argument names a *host group*, a list of services defined under `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Svchost`. DHCP Client's binary path is `svchost.exe -k LocalServiceNetworkRestricted -p`: it runs in the *LocalServiceNetworkRestricted* group as the low-privilege `NT AUTHORITY\LocalService` account, and `-p` applies additional process mitigation policies. On PCs with more than about 3.5 GB of RAM, Windows splits most group members into their own svchost instances, so you will often see one svchost per service in Task Manager.

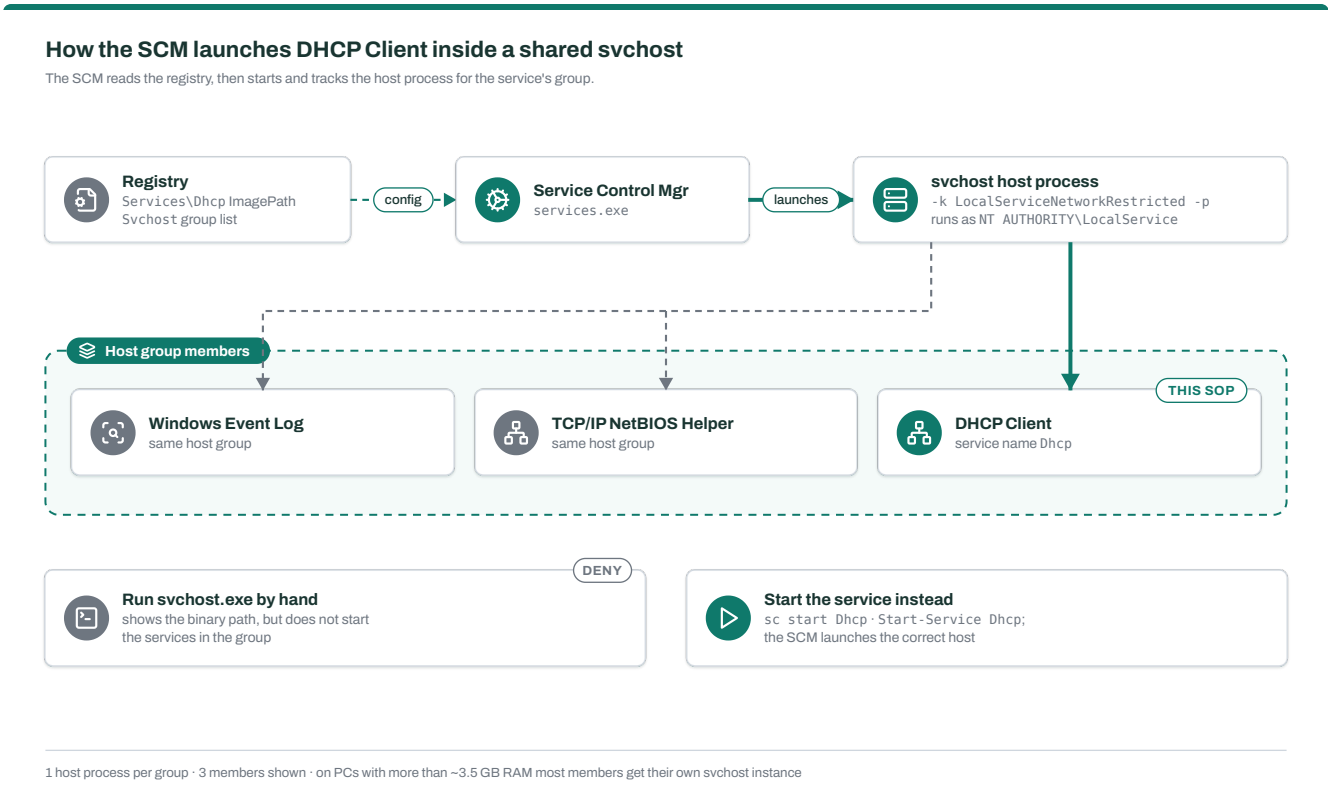


Figure 3. Start a hosted service through the SCM, never by launching svchost yourself, so the right host and account are used.

CORRECTION · RUNNING SVCHOST.EXE BY HAND

The field notes from the reference engagement include running `svchost.exe -k LocalServiceNetworkRestricted -p` from an elevated prompt. That command is useful for *reading* (it is exactly the service's binary path), but launching svchost manually does not start the services in the group: svchost is designed to be started and tracked by the SCM. To start the service, use `sc start Dhcp`, `net start Dhcp`, `Start-Service Dhcp` or services.msc, which make the SCM launch the correct host.

1.3 Service dependencies

A **dependency** tells the SCM "start these first, and do not start me if they fail". Dependencies are stored in the `DependOnService` (and, less commonly, `DependOnGroup`) values of the service's registry key. On current Windows 10 and Windows 11 builds, DHCP Client depends on three components:

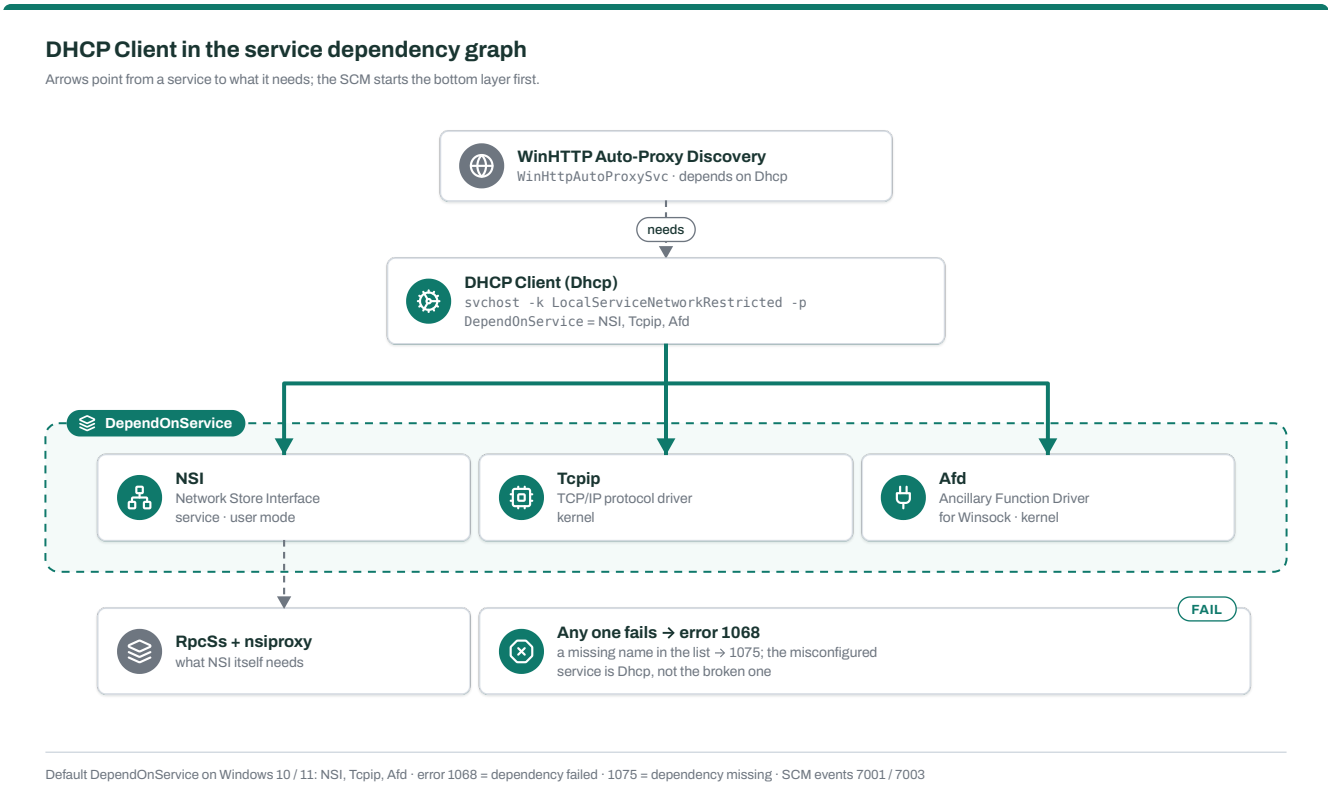


Figure 4. An error 1068 or 1075 means the dependency list of DHCP Client is wrong, even though the service named in the error is healthy.

Name	Type	Role
NSI	Service (Network Store Interface Service)	Delivers network configuration and change notifications to user-mode components. If it is stopped, network status and configuration break widely.

Tcpip	Kernel driver	The TCP/IP protocol stack itself.
Afd	Kernel driver (Ancillary Function Driver for Winsock)	The kernel side of Winsock; every socket a user-mode program opens goes through it.

If any listed dependency fails to start, the SCM refuses to start DHCP Client and logs **error 1068** ("The dependency service or group failed to start"). If the list names something that does not exist (for example a component left behind by uninstalled VPN or security software, or a mistyped manual edit), the error is **1075** ("The dependency service does not exist or has been marked for deletion"). Either way, the service whose configuration is wrong is not the one that is broken.

WHY · THE RECOVERY IN THE REFERENCE ENGAGEMENT WORKED

The field fix was `sc config dhcp depend= /`, which **clears** DHCP Client's dependency list. With nothing to wait for, the SCM started the service, the workstation leased an address, and the rest of the relay race completed: domain logon, Group Policy and mapped drives all returned. It worked because the actual network stack (Tcpip, Afd) was healthy and the *dependency entry* was what had gone wrong. It is a valid recovery technique, but it removes a safety rail: DHCP Client no longer waits for the stack at boot. The durable fix is to restore the default list, `NSI/Tcpip/Afd`, once you know why it broke (Chapter 3).

1.4 Secure channel, Group Policy and the event logs

- **Secure channel**: every domain member has a computer account in Active Directory with a password the machine changes automatically (every 30 days by default). Netlogon uses that shared secret to build an authenticated channel to a domain controller. If the local copy and the AD copy disagree (restored snapshot, duplicate name, long time offline with a reset account), domain logons fail with a trust-relationship error even though the network is fine.
- **Group Policy** applies computer settings at startup and user settings at logon, then refreshes every 90 minutes (±30). `gpupdate /force` reapplies everything now; `gpresult /r` shows what applied and from which domain controller.
- **Event logs**: the System log records SCM, Netlogon and Group Policy failures. The *Service Control Manager* event IDs 7000, 7001, 7003, 7023 and 7024 describe start failures and name the dependency that failed; Netlogon 5719 means no domain controller could be reached; GroupPolicy 1129 means policy processing failed for lack of network connectivity to a domain controller.

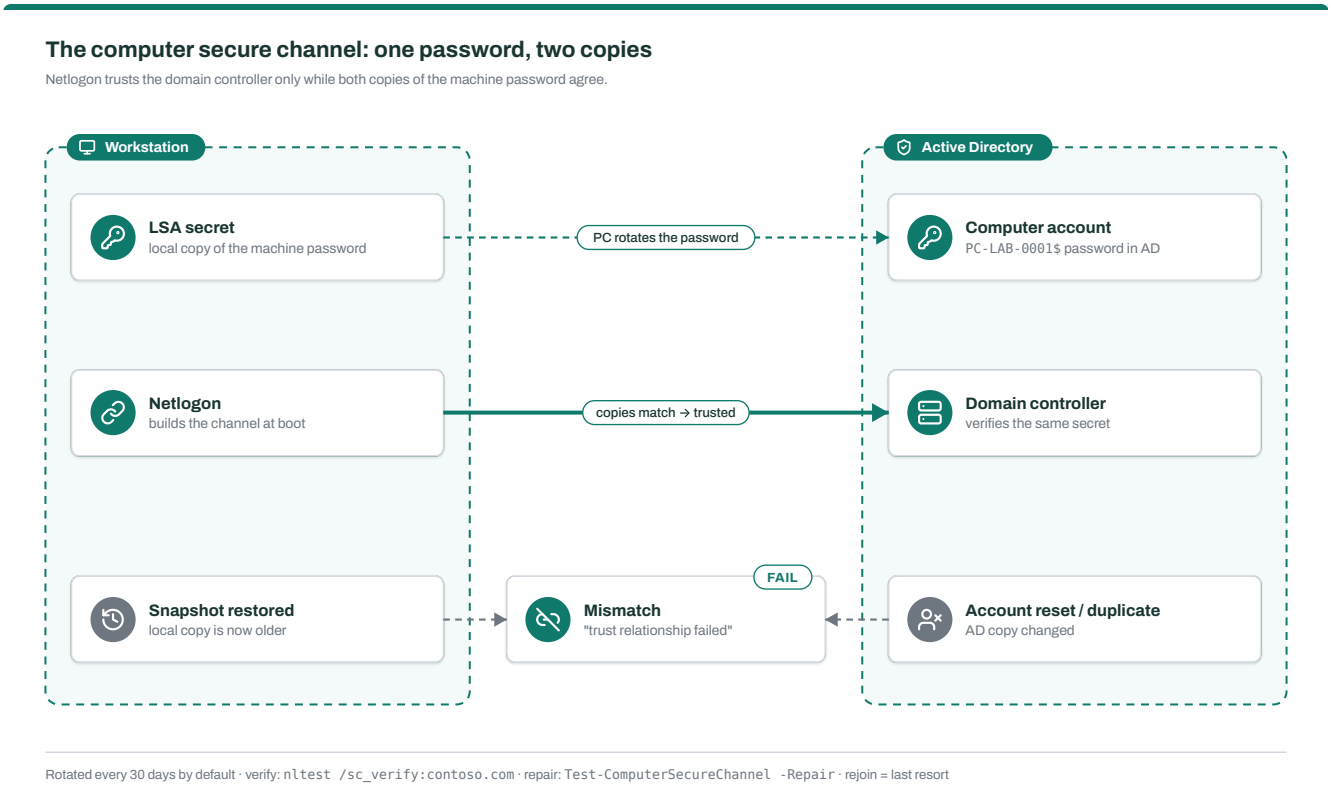


Figure 5. A trust failure can happen on a perfectly healthy network, so examine it only after the workstation has a valid address.

1.5 Choosing a method

A · Consoles (GUI)

services.msc, the Dependencies tab, Event Viewer and network settings.

BEST FOR: DIAGNOSIS AT THE DESK

B · Command line

sc, ipconfig, nltest and gpupdate from an elevated prompt.

BEST FOR: FAST, EXACT REPAIR

C · PowerShell

Get-Service, Test-ComputerSecureChannel and a repair template with evidence.

BEST FOR: REPEATABLE, DOCUMENTED FIXES

D · Domain rejoin

Remove and re-add the computer when the secure channel cannot be repaired.

BEST FOR: LAST RESORT FOR TRUST FAILURES

E · Enterprise

Intune Remediation and Group Policy to detect and prevent configuration drift.

BEST FOR: FLEETS, PREVENTION

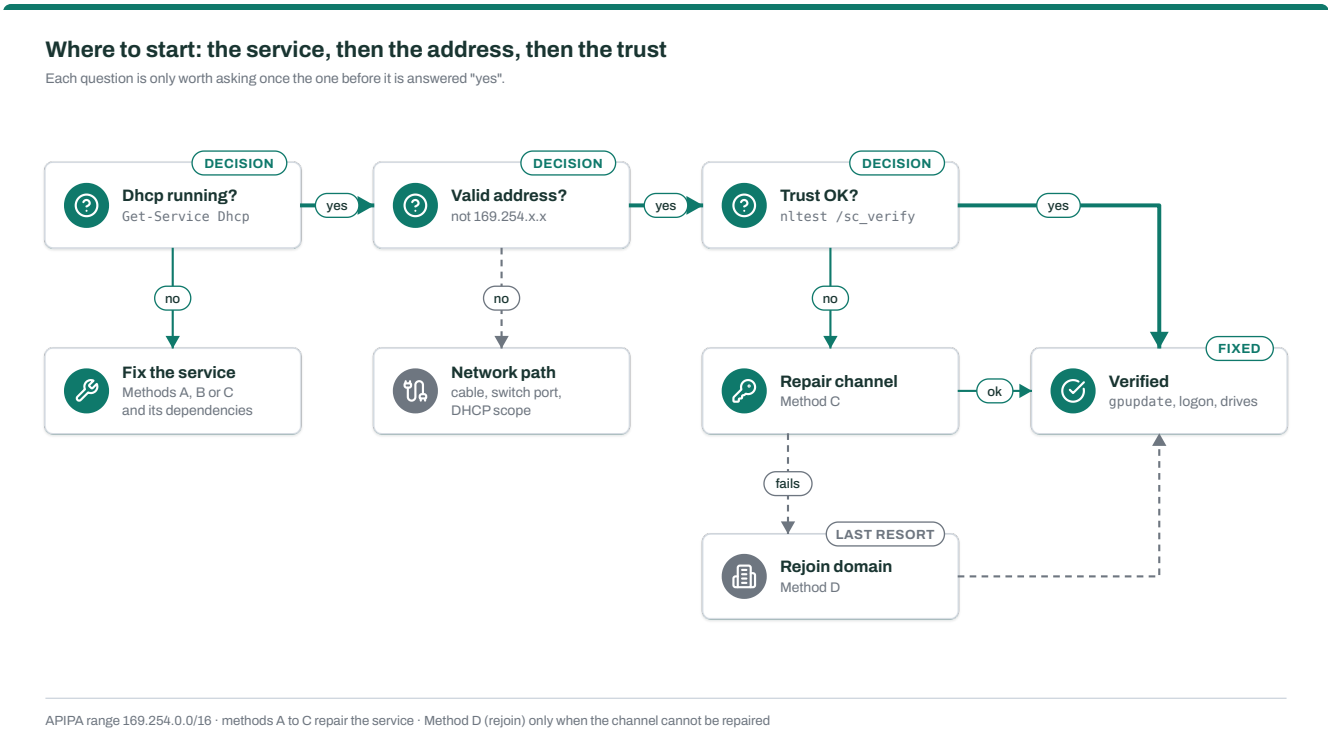


Figure 6. Work through the service, the address and the trust in that order; rejoining the domain is the last resort.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Task	Minimum access	Notes
View services, dependencies, IP configuration	Standard user	`sc qc`, `ipconfig /all` and `Get-Service` work unelevated.
Read the System event log	Standard user locally; *Event Log Readers* group remotely	Security log needs administrator rights.
Start/stop services, change dependencies (`sc config`)	Local Administrators group, elevated	The workstation may be unable to validate domain credentials, so plan for a local account.
Sign in while the domain is unreachable	Windows LAPS local administrator, or cached domain credentials	Retrieve the LAPS password from AD or Entra ID *before* any unjoin; it cannot be retrieved for a machine that never reconnects.
Repair the secure channel	Local admin + a domain account with *Reset password* rights on the computer object	Typically delegated to the support team on the workstation OUs, not Domain Admins.
Rejoin the domain	Local admin + delegated *Create/Delete Computer objects* on the target OU	Many hardened domains set `ms-DS-MachineAccountQuota` to 0, so ordinary users cannot join machines.
Intune Remediations	Intune Administrator or a custom role with *Device configurations* permissions	Requires Windows Enterprise/Education E3/E5 (or equivalent) device licensing.
Group Policy for service settings	Edit rights on the GPO and link rights on the OU	Group Policy Creator Owners or delegated GPO editors.

GOOD PRACTICE · LEAST PRIVILEGE FOR TRUST REPAIR

Do not use a Domain Admin account on a workstation whose state you do not trust. Delegate *Reset password* and *Validated write to service principal name / DNS host name* on workstation OUs to a dedicated support group, and activate membership just-in-time where your identity platform supports it.

2.2 Prerequisites checklist

- [ ] Ticket or change record; note the reported symptoms in order of appearance.
- [ ] Physical or console access to the workstation (remote tools will not work while it has no address).
- [ ] Local administrator credential (LAPS) retrieved and tested.
- [ ] Domain credential with delegated rights, for Methods C and D.
- [ ] A known-good workstation of the same Windows build, to compare service configuration if needed.
- [ ] Confirmation the network port works (another device on the same cable or dock gets an address).

2.3 Risk and planning notes

Action	Risk	Notes
`sc qc`, `ipconfig /all`, `nltest /sc_query`	READ-ONLY	Always safe; do these first.
Restore default dependencies	LOW	Returns the service to its shipped configuration.
Clear dependencies (`depend= /`)	MODERATE	Temporary recovery only; record it and restore defaults afterwards.
Secure-channel repair	MODERATE	Resets the computer account password; harmless if it was already in sync.
Domain rejoin	HIGHER	Needs restarts and local credentials; group memberships survive only if the existing computer object is reused.

CHAPTER 3

Methods A and B · Consoles and the Command Line

Methods A and B are presented together because a support engineer normally switches between them: the consoles to *see* the problem, the command line to *fix* it precisely. The steps follow the reference engagement in a safer order: diagnose first, restore the correct configuration, and use the "clear all dependencies" technique only when needed.

STEP 1 Confirm the symptom

1. Boot normally and sign in with the LAPS local administrator account (or cached domain credentials).
2. Open an elevated Command Prompt and run `ipconfig /all`. A connected Ethernet or Wi-Fi adapter with no IPv4 address, or *DHCP Enabled: Yes* with no lease details, points to the DHCP Client service.
3. Press Win+R, run `services.msc` and find **DHCP Client**. Note its **Status** and **Startup Type** (default: Automatic).
4. Try **Start**. If Windows reports error 1068 or 1075, open **Properties › Dependencies**: the top box lists what DHCP Client depends on.
5. Open `eventvwr.msc` › **Windows Logs › System**, filter by source *Service Control Manager*. Event 7001 or 7003 names the dependency that failed or is missing.

STEP 2 Inspect configuration and dependencies

ELEVATED COMMAND PROMPT · INSPECT — CMD

```
sc qc Dhcp
sc query Dhcp
sc query NSI
sc query Tcpip
sc query Afd
sc enumdepend Dhcp          &:: which services depend on DHCP Client
```

On a healthy Windows 10/11 machine, `sc qc Dhcp` returns output like this:

EXPECTED SC QC OUTPUT (HEALTHY) — TEXT

```
SERVICE_NAME: Dhcp
        TYPE               : 20  WIN32_SHARE_PROCESS
        START_TYPE         : 2   AUTO_START
```

```
        ERROR_CONTROL      : 1   NORMAL
        BINARY_PATH_NAME   : C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p
        LOAD_ORDER_GROUP   : TDI
        DISPLAY_NAME       : DHCP Client
        DEPENDENCIES       : NSI
                           : Tcpip
                           : Afd
        SERVICE_START_NAME : NT Authority\LocalService
```

Compare line by line. Typical faults: an extra dependency name (1075), a dependency whose own `sc query` shows STOPPED or a start failure (1068), a start type of DISABLED, or a changed service account.

STEP 3 Repair, in order of preference

1. **Fix the failing dependency** if one is genuinely stopped. For NSI: `sc config NSI start= auto` then `sc start NSI`. Tcpip or Afd failing to load indicates a damaged network stack: see Chapter 6.
2. **Restore the default dependency list** if it contains wrong or extra entries. The separator is a forward slash, and the space after `depend=` is mandatory.
3. **Clear the list as a temporary recovery** only if you need the machine online now (for example to pull a fix from the network) and the entries cannot be corrected yet. Restore the defaults before closing the ticket.
4. Make sure the start type is automatic, then start the service.

ELEVATED COMMAND PROMPT · REPAIR CMD

```
:: Preferred: restore the shipped dependencies (Windows 10/11)
sc config Dhcp depend= NSI/Tcpip/Afd

:: Temporary recovery only: clear ALL dependencies, then restore later
sc config Dhcp depend= /

:: Start type and start
sc config Dhcp start= auto
sc start Dhcp
sc query Dhcp
```

CAUTION · WHAT "DEPEND= /" REALLY DOES

The command writes an empty `DependOnService` value. DHCP Client will then start as early as the SCM chooses, with no guarantee that TCP/IP and Winsock are ready, and the SCM will no longer stop it automatically if they fail. Most of the time it still works, which is why it is a popular fix, but it hides the real fault and leaves a non-standard build. Record it in the ticket and restore `NSI/Tcpip/Afd` after the root cause is removed.

STEP 4 Re-establish the network and the domain

ELEVATED COMMAND PROMPT · NETWORK AND DOMAIN CMD

```
ipconfig /renew
ipconfig /all                       &:: address, gateway, DNS servers, lease time
ipconfig /registerdns               &:: refresh this PC's DNS records
nltest /dsgetdc:contoso.com         &:: can we locate a domain controller?
nltest /sc_query:contoso.com        &:: state of the secure channel
nltest /sc_verify:contoso.com       &:: verify it (and the trust password)
gpupdate /force
gpresult /r /scope computer
```

1. Confirm a valid address, gateway and the corporate DNS servers.
2. `nltest /dsgetdc` must return a DC name. If it fails, the problem is DNS or reachability, not the trust.
3. `nltest /sc_verify` should end in *Trust Verification Status = 0 0x0 NERR_Success*. If not, repair the secure channel (Method C) or rejoin (Method D).
4. `gpupdate /force` should report both computer and user policy updated successfully.
5. Restart the workstation and confirm DHCP Client starts automatically, the domain logon works and mapped drives appear.

CHAPTER 4

Method C · PowerShell

4.1 Inspecting services the PowerShell way

PowerShell returns services as objects, so dependencies, dependents and start types can be compared and logged. Two cautions: in PowerShell `sc` is an alias for `Set-Content`, so always type `sc.exe`; and `Set-Service` cannot change dependencies, so dependencies are changed with `sc.exe` or in the registry.

INSPECT (READ-ONLY) POWERSHELL

```
Get-Service Dhcp | Select-Object Name, Status, StartType
Get-Service Dhcp -RequiredServices      # what DHCP Client depends on
Get-Service Dhcp -DependentServices     # what depends on DHCP Client
(Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\Dhcp).DependOnService
Get-CimInstance Win32_Service -Filter "Name='Dhcp'" | Select-Object Name, State, StartMode, StartName,
PathName

#  Recent start failures and the dependency they name
Get-WinEvent -FilterHashtable @{ LogName = 'System'; ProviderName = 'Service Control Manager'
                                 Id = 7000,7001,7003,7023,7024; StartTime = (Get-Date).AddDays(-2) } |
  Select-Object TimeCreated, Id, Message | Format-List

Get-NetIPConfiguration | Select-Object InterfaceAlias, IPv4Address, IPv4DefaultGateway, DNSServer
```

4.2 Secure-channel repair without rejoining

When the address and DNS are fine but domain authentication fails, repair the secure channel first. It resets the computer account password on both sides, keeps the computer object, its group memberships and its SID, and usually needs no restart. Both cmdlets ship with **Windows PowerShell 5.1**; run them there rather than in PowerShell 7.

WINDOWS POWERSHELL 5.1 · ELEVATED POWERSHELL

```
Test-ComputerSecureChannel -Verbose                        # True = healthy
Test-ComputerSecureChannel -Repair -Credential (Get-Credential CONTOSO\svc-deskside) -Verbose
#  Alternative: target a specific DC (useful in multi-site domains)
Reset-ComputerMachinePassword -Server DC01.contoso.com -Credential (Get-Credential CONTOSO\svc-deskside)
```

4.3 Base template · DHCP Client and domain recovery with evidence

REPAIR-DHCPDOMAIN.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Restore DHCP Client (Dhcp) configuration and domain connectivity on Windows 10/11.
.NOTES     Run elevated in Windows PowerShell 5.1 (secure-channel cmdlets are 5.1-only).
           Default dependencies below match current Windows 10/11 builds; confirm against a healthy
reference PC.
#>
[CmdletBinding(SupportsShouldProcess)]
```

```
param(
  [string]   $DomainName = 'contoso.com',
  [string[]] $DefaultDependencies = @('NSI','Tcpip','Afd'),
  [switch]   $ClearDependencies,             # temporary recovery: start with no dependencies
  [switch]   $RepairSecureChannel,
  [pscredential] $DomainCredential,          # account with Reset password on the computer object
  [string]   $EvidencePath = "$env:PUBLIC\Documents\DhcpDomain-$(Get-Date -f yyyyMMdd-HHmm).json"
)
$ErrorActionPreference = 'Stop'
$key = 'HKLM:\SYSTEM\CurrentControlSet\Services\Dhcp'
$ev  = [ordered]@{ Computer = $env:COMPUTERNAME; Operator = "$env:USERDOMAIN\$env:USERNAME"; StartUtc =
(Get-Date).ToUniversalTime() }

# ---- Stage 1: capture current state (read-only) --------------------------
$ev.DependsBefore = @((Get-ItemProperty $key).DependOnService)
$ev.StatusBefore  = (Get-Service Dhcp).Status.ToString()
$ev.StartBefore   = (Get-Service Dhcp).StartType.ToString()
& reg.exe export 'HKLM\SYSTEM\CurrentControlSet\Services\Dhcp' "$env:PUBLIC\Documents\Dhcp-backup.reg" /y
| Out-Null
foreach ($d in $DefaultDependencies) { $ev["Dep_$d"] = (Get-Service $d -ErrorAction
SilentlyContinue).Status }

# ---- Stage 2: dependencies -----------------------------------------------
$target = if ($ClearDependencies) { '/' } else { $DefaultDependencies -join '/' }
$wrong  = (($ev.DependsBefore -join '/') -ne ($DefaultDependencies -join '/')) -or $ClearDependencies
if ($wrong -and $PSCmdlet.ShouldProcess('Dhcp', "sc.exe config depend= $target")) {
  & sc.exe config Dhcp depend= $target | Out-Null
  $ev.DependsSetTo = $target
}

# ---- Stage 3: start type and start ---------------------------------------
if ($PSCmdlet.ShouldProcess('Dhcp', 'Set Automatic and start')) {
  Set-Service -Name Dhcp -StartupType Automatic
  if ((Get-Service Dhcp).Status -ne 'Running') { Start-Service Dhcp }
  Start-Sleep -Seconds 10
  & ipconfig.exe /renew | Out-Null
}
$ev.IPv4 = @(Get-NetIPAddress -AddressFamily IPv4 -ErrorAction SilentlyContinue |
             Where-Object { $_.PrefixOrigin -eq 'Dhcp' } | ForEach-Object IPAddress)

# ---- Stage 4: domain controller and secure channel -----------------------
$ev.DcLocator     = (& nltest.exe "/dsgetdc:$DomainName" 2>&1 | Select-Object -First 1) -join ''
$ev.SecureChannel = Test-ComputerSecureChannel -ErrorAction SilentlyContinue
if (-not $ev.SecureChannel -and $RepairSecureChannel -and $PSCmdlet.ShouldProcess($DomainName, 'Repair
secure channel')) {
  if (-not $DomainCredential) { $DomainCredential = Get-Credential -Message 'Account with Reset password
rights' }
  $ev.SecureChannelRepaired = Test-ComputerSecureChannel -Repair -Credential $DomainCredential
}

# ---- Stage 5: policy refresh and evidence --------------------------------
if ($ev.SecureChannel -or $ev.SecureChannelRepaired) { & gpupdate.exe /target:computer /force | Out-Null
}
$ev.StatusAfter = (Get-Service Dhcp).Status.ToString()
$ev.EndUtc      = (Get-Date).ToUniversalTime()
$ev | ConvertTo-Json -Depth 3 | Out-File $EvidencePath -Encoding utf8
Write-Host "DHCP Client: $($ev.StatusAfter); IPv4: $($ev.IPv4 -join ', '); evidence: $EvidencePath"
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION POWERSHELL

```
#  Preview what would change
.\Repair-DhcpDomain.ps1 -DomainName contoso.com -RepairSecureChannel -WhatIf
#  Restore defaults, start the service, repair the trust if needed
.\Repair-DhcpDomain.ps1 -DomainName contoso.com -RepairSecureChannel -DomainCredential (Get-
Credential CONTOSO\svc-deskside)
```

CHAPTER 5

Methods D and E · Rejoin and Enterprise Prevention

5.1 Method D · Rejoining the domain

Rejoin only when the secure channel cannot be repaired: for example when the computer object was deleted, the repair account is refused, or `Test-ComputerSecureChannel -Repair` keeps returning False. A rejoin is slower and riskier than a repair, but it is a clean slate.

1. Retrieve and test the local administrator password (LAPS) *before* unjoining. After unjoin, domain credentials will not work on this PC.
2. In AD, check whether the computer object still exists and note its OU and group memberships. Reusing the existing object (same name) preserves them.
3. **GUI**: Win + R › `sysdm.cpl` › **Computer Name** › **Change...** › select **Workgroup** (e.g. WORKGROUP) › restart, sign in with the local account, then return and select **Domain**, enter `contoso.com` and the delegated credential › restart.
4. **PowerShell**: use the commands below (Windows PowerShell 5.1).
5. After the final restart, sign in with a domain account, run `gpupdate /force`, and confirm drives and policies.

WINDOWS POWERSHELL 5.1 · REJOIN — POWERSHELL

```
#  1. Leave the domain (uses the domain credential if the DC is reachable; -Force skips the confirmation)
Remove-Computer -UnjoinDomainCredential (Get-Credential CONTOSO\svc-deskside) -WorkgroupName WORKGROUP -
Restart -Force

#  2. After restart, signed in as the local administrator:
Add-Computer -DomainName contoso.com -Credential (Get-Credential CONTOSO\svc-deskside) `
  -OUPath 'OU=Workstations,DC=contoso,DC=com' -Restart
```

NOTE · OFFLINE DOMAIN JOIN

When the machine cannot reach a domain controller at all, an administrator can pre-create the account on a domain-connected server with `djoin /provision` and apply it on the workstation with `djoin /requestodj`. The join completes at next boot when a DC becomes reachable.

5.2 Method E · Remote help once the network is back

A workstation without an address cannot be reached remotely, so the first fix is always local. Once it has an address, remote PowerShell can verify and finish the job, and can

sweep other machines for the same drift before they reboot into the same failure.

REMOTE VERIFICATION AND FLEET SWEEP — POWERSHELL

```powershell
Invoke-Command -ComputerName PC-LAB-0001 -ScriptBlock {
  [pscustomobject]@{ Dhcp = (Get-Service Dhcp).Status
                     Depends = (Get-ItemProperty
HKLM:\SYSTEM\CurrentControlSet\Services\Dhcp).DependOnService -join '/'
                     Trust = Test-ComputerSecureChannel }
}
#  Find machines whose dependency list has drifted (takes effect at their NEXT boot)
$pcs = Get-ADComputer -Filter * -SearchBase 'OU=Workstations,DC=contoso,DC=com' | ForEach-Object Name
Invoke-Command -ComputerName $pcs -ErrorAction SilentlyContinue -ScriptBlock {
  (Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\Dhcp).DependOnService -join '/'
} | Where-Object { $_ -ne 'NSI/Tcpip/Afd' }
```

5.3 Method E · Intune Remediation for configuration drift

KEY CONCEPT · CONFIGURATION IS READ AT START TIME

The SCM reads a service's dependencies only when it starts the service. A machine whose dependency list was damaged today will keep working until its next restart, then fail. That gap is the opportunity for proactive detection: find the drift while the device is still online and fix it before the reboot.

Create the pair in **Intune admin center › Devices › Scripts and remediations › Remediations › + Create**; run as SYSTEM in 64-bit PowerShell, schedule daily, and assign to a Windows device group.

DETECT-DHCPDEPENDENCIES.PS1 · DETECTION — POWERSHELL

```powershell
$expected = 'NSI/Tcpip/Afd'
$svc  = Get-Service Dhcp
$deps = ((Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\Dhcp).DependOnService) -join '/'
if ($deps -ne $expected -or $svc.StartType -ne 'Automatic') {
  Write-Output "Drift: depends='$deps' start=$($svc.StartType)"; exit 1
}
Write-Output 'DHCP Client configuration OK'; exit 0
```

REMEDIATE-DHCPDEPENDENCIES.PS1 · REMEDIATION — POWERSHELL

```powershell
& sc.exe config Dhcp depend= NSI/Tcpip/Afd | Out-Null
Set-Service -Name Dhcp -StartupType Automatic
if ((Get-Service Dhcp).Status -ne 'Running') { Start-Service Dhcp }
Write-Output 'DHCP Client dependencies restored to NSI/Tcpip/Afd'; exit 0
```

WATCH OUT · VALIDATE THE BASELINE PER BUILD

Hard-code the expected list only after confirming it on each Windows build in your estate (`sc qc Dhcp` on a clean reference machine). If a future build changes the defaults, a stale remediation would itself become the source of drift.

5.4 Method E · Group Policy

- **Computer Configuration › Policies › Windows Settings › Security Settings › System Services › DHCP Client**: define the startup mode as *Automatic* so a local change to Disabled is reverted at the next policy refresh.
- A Group Policy Preferences *Registry* item can enforce `DependOnService` (REG_MULTI_SZ) on `HKLM\SYSTEM\CurrentControlSet\Services\Dhcp` if your organisation prefers GPO to Intune. Use item-level targeting by OS build.

CHAPTER 6

Verification, Rollback and Troubleshooting

6.1 Verification checklist

Service and network

- [ ] `sc qc Dhcp` shows NSI, Tcpip, Afd and AUTO_START
- [ ] DHCP Client running after a full restart
- [ ] Valid IPv4 lease, gateway and corporate DNS
- [ ] `nslookup contoso.com` resolves domain controllers
- [ ] No new SCM 7000-series errors in the System log

Domain and user

- [ ] `nltest /sc_verify` returns NERR_Success
- [ ] `gpupdate /force` completes for computer and user
- [ ] A domain user signs in without cached-logon warnings
- [ ] Mapped drives and printers appear
- [ ] User confirms normal operation

Attach the template's JSON evidence, the before/after `sc qc Dhcp` output and the exported `Dhcp-backup.reg` to the ticket.

6.2 Rollback

Change	How to undo it
Dependencies changed	Import the exported `Dhcp-backup.reg` (`reg import`) or run `sc config Dhcp depend= <original list>`; restart the service.
Dependencies cleared	`sc config Dhcp depend= NSI/Tcpip/Afd`. This is the expected follow-up, not just a rollback.
Secure channel reset	Nothing to undo; the new password is now in sync on both sides.
Domain rejoin	Re-add group memberships if a new computer object was created; remove any stale duplicate object.

6.3 Troubleshooting flowchart

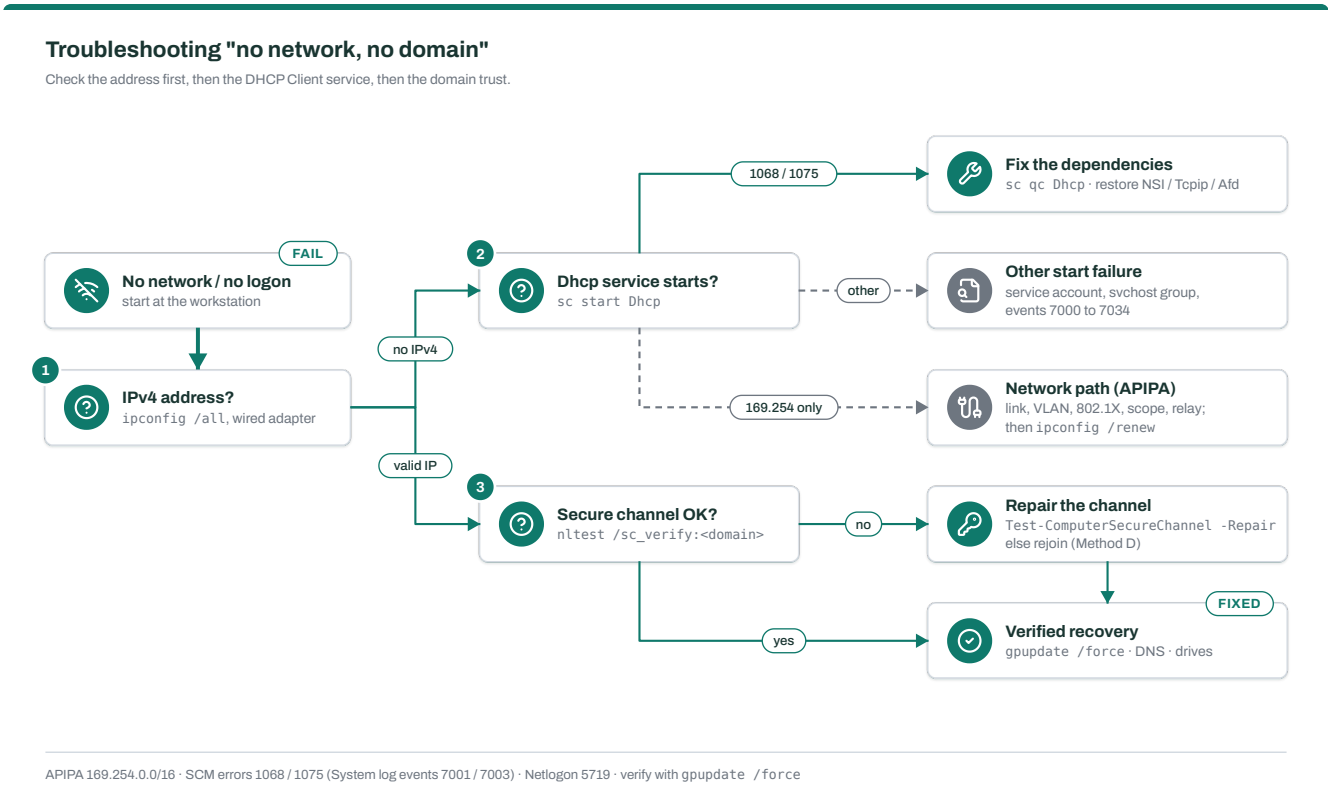


Figure 7. Each branch ends in a specific fix, and every successful path finishes with the same verification step.

6.4 Symptom, cause and fix

Symptom	Likely cause and fix
Error 1068 starting DHCP Client	A dependency failed to start. `sc query` NSI, Tcpip, Afd; fix the failing one, or restore defaults if the list is wrong.
Error 1075	The list names a component that does not exist (leftover from uninstalled software). Restore `NSI/Tcpip/Afd`.
Error 1079 or logon failure (1069)	The service account was changed. DHCP Client must run as `NT Authority\LocalService`: `sc config Dhcp obj= "NT Authority\LocalService" password= ""`.
Tcpip or Afd will not load	Damaged network stack or a faulty third-party filter driver. `netsh winsock reset` and `netsh int ip reset`, remove recently installed VPN/security software, then DISM and SFC.
169.254.x.x address	Service fine, no server reply: cable, switch port, VLAN, 802.1X authentication, scope exhausted or relay (IP helper) misconfigured.

"No logon servers available"	DNS points to the wrong servers or DC unreachable. `nltest /dsgetdc`, `nslookup -type=SRV _ldap._tcp.dc._msdcs.contoso.com`.
"Trust relationship … failed"	Secure channel broken. `Test-ComputerSecureChannel -Repair`; rejoin if that fails. Check for duplicate computer names or restored snapshots.
Problem returns after every reboot	Something re-applies the bad configuration: a GPO, a configuration-management package or security software. Find it with `gpresult /h` and software inventory before fixing again.

CHAPTER 7

Summary, Glossary and Review

7.1 Chapter summary

- Domain logon is a chain: link, DHCP, DNS, DC locator, secure channel, Group Policy, drives. Fix the first broken link.
- No IPv4 address at all means the DHCP Client service is not running; 169.254.x.x means it runs but no server answered.
- DHCP Client depends on NSI, Tcpip and Afd. Errors 1068 and 1075 point at the dependency list, not at DHCP itself.
- `sc config Dhcp depend= /` is a temporary recovery; restore `NSI/Tcpip/Afd` afterwards.
- Repair the secure channel before rejoining; rejoin only as a last resort, with the LAPS password in hand.

Check your understanding

1. A user reports no internet and missing mapped drives; `ipconfig` shows no IPv4 address on a connected adapter. Which service do you check first, and why are the drives missing?
2. What is the difference between error 1068 and error 1075, and what does each tell you to look at?
3. Write the command that restores DHCP Client's default dependencies, and explain why the space after `depend=` matters.
4. Why is launching `svchost.exe -k LocalServiceNetworkRestricted -p` by hand not a way to start DHCP Client?
5. When would you choose `Test-ComputerSecureChannel -Repair` over a domain rejoin, and what must you have before a rejoin?

7.2 Glossary

DHCP / DORA	Protocol that leases IP settings; Discover, Offer, Request, Acknowledge.
APIPA	Automatic Private IP Addressing (169.254.x.x) when no DHCP server replies.
SCM	Service Control Manager (`services.exe`), which starts and supervises services.
svchost host group	Named set of services hosted by `svchost.exe -k <group>`.
Service dependency	A service or driver that must start before another (`DependOnService`).

NSI / Tcpip / Afd	Network Store Interface service, TCP/IP driver, Winsock kernel driver.
Secure channel	Authenticated Netlogon channel between a domain member and a DC.
LAPS	Local Administrator Password Solution; managed, rotated local admin passwords.